Corrections and Refinements

FFGFT / T0 Time-Mass Duality

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Contents

Preliminary Note

This document lists corrections and refinements for previously published documents in the FFGFT series. Older documents are *not* modified. Where a contradiction exists between an older and a newer document, the version recorded here is binding.

Each entry contains: affected document and location, the erroneous expression, the correct expression, and a brief justification.

1 Correction 1: Prefactor in the Higgs derivation of ξ

Affected Document

Doc. 049 (T0 Lagrangian and Higgs Integration), HTML draft version and earlier working versions.

Erroneous Expression

$$\xi = \frac{\lambda_h^2 v^2}{64\pi^4 m_h^2} \approx 1.0 \times 10^{-5} \qquad \text{(old – wrong)}$$

Correct Expression

$$\boxed{\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \approx 1.33 \times 10^{-4}} \tag{190.1}$$

with the experimental inputs:

$$m_h = 125.25\,\text{GeV} \quad \text{(Higgs mass)},$$
$$v = 246.22\,\text{GeV} \quad \text{(vacuum expectation value)},$$
$$\lambda_h = \frac{m_h^2}{2v^2} \approx 0.129 \quad \text{(Higgs self-coupling)}.$$

Justification

The prefactor $64\pi^4$ originated as a typo in the early HTML visualization. It yields $\xi \approx 10^{-5}$, which deviates by an order of magnitude from the geometrically derived value

$\xi = 4/3 \cdot 10^{-4}$ and does not reproduce the natural constants. The correct prefactor $16\pi^3$ yields $\xi \approx 1.33 \times 10^{-4}$, consistent with the value independently determined from the proton-electron mass ratio (Doc. 009).

2 Correction 2: Tau-Yukawa prefactor r_τ

Affected Document

Doc. 116 (Koide formula as a consequence of T0 theory), table of Yukawa coefficients.

Erroneous Expression

$$r_\tau = \frac{8}{3} \approx 2.667 \quad \Rightarrow \quad m_\tau \approx 1712\,\text{MeV} \quad (-3.6\,\%\ \text{dev.})$$

(old – wrong)

Correct Expression

$$\boxed{r_\tau = \frac{25}{9} \approx 2.778} \quad \Rightarrow \quad m_\tau \approx 1783\,\text{MeV} \quad (+0.4\,\%\ \text{dev.}) \tag{190.2}$$

The lepton masses are generally given by:

$$m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v \tag{190.3}$$

with exponents $p_e = 3/2$, $p_\mu = 1$, $p_\tau = 2/3$ (step size $\Delta p = 1/3$) and the vacuum expectation value v.

Justification

$r_\tau = 8/3$ was a remnant from an earlier iteration stage. Doc. 158 (Koide-to-g-2 bridge) already uses $r_\tau = 25/9$, which agrees numerically with the tau mass. Furthermore, $25/9 = (5/3)^2$ has a geometric interpretation as the square of the pure sixth in the natural overtone series, consistent with the musical resonance analogy (Doc. 060).

Complete Corrected Prefactor Table

Lepton	Exponent p	Prefactor r	m [MeV]
Electron	$3/2$	$4/3$	0.511
Muon	1	$16/5$	105.7
Tau	$2/3$	$25/9$	1783

3 Correction 3: Base formula for g-2 of the electron

Affected Document

T0 Explorer (HTML visualization), first version; earlier draft versions of Doc. 018.

Erroneous Expression

$$a_e = \frac{4\pi}{f \cdot k_{\text{geom}}} \qquad \text{(old – wrong)}$$

Correct Expression

$$\boxed{a_e = \frac{S_3}{f \cdot k_{\text{geom}}} = \frac{2\pi^2}{f \cdot k_{\text{geom}}}} \tag{190.4}$$

with $f = 1/\xi \approx 7500$, $k_{\text{geom}} = (2/\sqrt{\varphi}) \cdot \sqrt{2} \approx 2.224$ and $S_3 = 2\pi^2 \approx 19.74$ (surface area of the 4D unit sphere, i.e. the 3D boundary of the torus).

Justification

The earlier prefactor 4π corresponds to the surface area of the 2D sphere in 3D space, not the geometrically correct 3D surface of the 4D torus. $S_3 = 2\pi^2$ is the correct value from torus geometry. The fractal corrections for muon and tau still use 4π:

$$\Delta a_\mu = \frac{4\pi}{f^{5/3}}, \tag{190.5}$$

$$\Delta a_\tau = \frac{4\pi}{f^{4/3}}. \tag{190.6}$$

4 Refinement 1: Accuracy of the Koide formula

Affected Documents

Doc. 116 (abstract, line 14): $\Delta Q < 0.00003\,\%$
Doc. 158 (abstract): experimentally confirmed to $0.001\,\%$

Correct Interpretation

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{190.7}$$

is experimentally confirmed to better than $0.001\,\%$ (PDG values 2022). The figure $< 0.00003\,\%$ in Doc. 116 refers to the internal consistency of the T0 formulas, not the experimental measurement uncertainty. Both statements are substantively correct but refer to different quantities. In external communications, $0.001\,\%$ is the appropriate figure.

Note

The individual T0 masses agree with experiment to within $\sim 1\,\%$; the high precision of $Q = 2/3$ arises from the specific algebraic structure of the Koide combination, not from individual mass accuracy.

5 Refinement 2: $\hbar$ from the Higgs field – derivation chain

Context

Planck's constant $\hbar$ is derived differently in various documents. The complete, consistent derivation chain is fixed here.

Binding Derivation Chain

$$\xi = \frac{\lambda_h^2 v^2}{16\pi^3 m_h^2} \longrightarrow L_0 = \xi \cdot \ell_P \longrightarrow \hbar \sim \sqrt{\xi} \cdot \ell_P \cdot c \tag{190.8}$$

The time-mass binding condition provides the physical justification:

$$T(x,t)\cdot m(x,t) \;=\; 1 \quad\Longrightarrow\quad T \;=\; \frac{\hbar}{m\,c^2} \quad\Longrightarrow\quad E \;=\; \hbar\omega. \tag{190.9}$$

$h = 2\pi\hbar$ is therefore *not* a free constant but a geometric consequence of ξ and ℓ_P. The numerical SI value $h \approx 6.626\times10^{-34}$ J·s is the calibration of this relation onto the SI base units.

6 Refinement 3: Two ξ parameters – two distinct spaces

Context

In Documents 173–176 (quantum dots, spin qubits, qubit state spaces, Shor algorithm), two numerically different ξ values appear. Earlier documents did not separate these consistently, which can cause confusion.

Binding Distinction

Parameter	Value	Domain	Meaning
ξ_0 (geometric)	$\frac{4}{30000} \approx 1.33\times10^{-4}$	3D geometric space	Length hierarchy of the torus; valid in physical position space
ξ_{Higgs} (algebraic)	$\approx 1.04\times10^{-5}$	Number / quantum state space	Algebraic field corrections; Higgs sector; decoherence rates

Justification

$\xi_0 = 4/30000$ follows from the torus geometry of physical space (geometric derivation, independent of SI measurements). ξ_{Higgs} follows from the algebraic Higgs field structure and appears in coupling formulas between quantum states (e.g. Bell correlations, T_2 hierarchy).

The two ξ values are not a contradiction; they describe complementary aspects: geometry of space (ξ_0) vs. algebra of state space (ξ_{Higgs}). External communications must always specify which parameter is meant.

7 Refinement 4: L_0 and the Schwarzschild interpretation

Affected Documents

Earlier versions of Doc. 180–182 (Lagrangian derivation, torus geometry, maximum cosmological scale).

Erroneous Approach

Earlier versions interpreted $L_0 = \xi_0 \cdot \ell_P$ as the Schwarzschild radius of a fundamental minimal object. This interpretation was explicitly discarded in Doc. 182 (2026 revision).

Correct Interpretation

L_0 is exclusively the **geometric fundamental length** of the FFGFT torus:

$$L_0 = \xi_0 \cdot \ell_P \approx 2.15 \times 10^{-39}\,\text{m}. \tag{190.10}$$

It defines the lowest scale stage of the ξ hierarchy and has *no* Schwarzschild interpretation. The cosmological Hubble radius R_H is the system-specific upper limit of the cosmological sector – not a universal maximum scale. Each scale has its own system-specific upper limit.

Note: Schwarzschild as Consistency Check

The Schwarzschild connection remains valid as an **internal consistency check**: an object of mass $M_0 = \xi_0 \cdot m_P/2$ would have a Schwarzschild radius exactly equal to L_0 – with no free parameter. If M_0 matches a scale stage of the ξ hierarchy, that is a non-trivial self-consistency test. The connection is *a probe, not a derivation*.

8 Refinement 5: Scope and reach of the mass formulas

Context

Several FFGFT documents describe mass derivations for leptons and other particles. The scope of these derivations must be communicated clearly.

Binding Formulation

The FFGFT mass formulas (e.g. $m_\ell = r_\ell \cdot \xi^{p_\ell} \cdot v$) yield values that agree with experiment to $\sim 1\,\%$. This is **not a precision calculation** but a structural proof:

- The mass formulas *follow geometrically* from ξ and v – they are not fitted.
- The remaining $\sim 1\,\%$ deviations reflect measurement uncertainties and approximations, not errors in the structure.
- Numerical accuracy beyond the Standard Model is *not* claimed.

To be avoided in external texts: FFGFT calculates the masses exactly. The correct statement is: FFGFT *derives* the mass structure from a single parameter ξ and reproduces the orders of magnitude to $\sim 1\,\%$.

9 Refinement 6: Koide formula only with original values

Binding Rule

The Koide relation

$$Q = \frac{m_e + m_\mu + m_\tau}{(\sqrt{m_e} + \sqrt{m_\mu} + \sqrt{m_\tau})^2} = \frac{2}{3} \tag{190.11}$$

must be evaluated only with experimentally measured masses (PDG values).

Not permitted: substituting the symbolic FFGFT terms $r_\ell \cdot \xi^{p_\ell} \cdot v$ directly into the formula. The $\sqrt{m_i}$ operation generates cross-terms with mixed ξ powers that are foreign to the torus structure space. Any such substitution yields $Q \approx 0.6677 \neq 2/3$ and is not a contradiction to the theory but a category error.

Permitted: evaluating FFGFT terms numerically (e.g. $m_\tau = 1783$ MeV) and inserting that result into the Koide formula like a measured value.

Mathematical background: The r_ℓ terms are geometric invariants of their torus modes. The set of admissible pairs (r_i, p_i) is not closed under multiplication: $r_e \cdot r_\mu = 64/15$ does not belong to any torus mode. This is the Non-Closure Theorem (Doc. 193). The general analysis of composition limits is found in Doc. 192.

10 Refinement 7: Terminology renormalization group and scale structure

Affected Documents

- Doc. 005 (T0 tm-extension): RG flow, renormalization group factor, renormalization group invariance, subsection renormalization group treatment.
- Doc. 008 (ξ and e): subsection modified renormalization group equations.
- Doc. 019 (T0 Lagrangian, earlier version): subsection renormalization group equations.
- Doc. 086 (document overview): renormalization group as flow in parameter space.
- Doc. 189 (torus derivation): cumulative renormalization group running.
- Doc. 202 (FFGFT comprehensive field theory), §18: section title The renormalization group.

Previous Formulation

In the documents listed, the ξ_0-modified scale running of the coupling

$$\frac{d\alpha}{d\ln\mu} = \beta_0\,\alpha^2\Big(1+\xi_0\,\ln\tfrac{\mu}{E_0}\Big) \qquad \text{(old – imprecise)}$$

and the corresponding scale dependence are designated as the *renormalization group*, sometimes adjacent to statements explicitly characterizing ξ_0 as a topologically fixed geometric constant.

Refined Formulation

In FFGFT, the scale dependence of the coupling is *not* a result of a renormalization procedure but a direct consequence of the recursion $\mathcal{R}$ (Doc. 203) on the fractal torus geometry. The correct designation is therefore:

$$\boxed{\begin{aligned}&\text{Scale structure from recursion}\\ &\quad\equiv\ \text{recursive } \alpha \text{ running via } \mathcal{R}\end{aligned}} \tag{190.12}$$

Specifically:

- The term $\xi_0\,\ln(\mu/E_0)$ arises from the iterated application of the recursion $\mathcal{R}$ to the mode structure, not from a counter-term subtraction.
- ξ_0 is *not* a fixed point of a β function but a topologically fixed constant ($\xi_0 = 4/30000$, Doc. 180).
- Stability, scale flow, and UV finiteness are three aspects of *the same* recursive structure – not separate mechanisms.

Language Convention

Avoid	Use
Renormalization group (implies QFT procedure)	Scale structure from recursion
RG flow / RG running	Recursive α running; $\mathcal{R}$-induced scale correction
Renormalization group equations	Scale equations from $\mathcal{R}$
Renormalization group invariance	Recursive scale invariance
Renormalization group fixed point	Recursion fixed point: $\mathcal{R}(\Phi_*) = \Phi_*$

Justification

The designation renormalization group imports a tool of standard quantum field theory based on a fundamentally different principle: there UV divergences are absorbed by a procedure, and parameters *run* under a formal scale transformation. In FFGFT neither the divergence problem (cf. Doc. 159, without artificial renormalization) nor a free scale parameter to be absorbed exists. The scale dependence is a *structural property* of the recursion, not a subsequent adjustment step.

The FFGFT-native view is already correctly formulated in Doc. 159 (renormalization as symptom, not as solution) and Doc. 189 (geometrically determined, not by renormalization); this clarification serves to unify the terminology across the older Lagrangian and ξ-derivation series (Doc. 005, 008, 019) and across Doc. 202.

The underlying mechanism is developed in Doc. 203 (recursion operator $\mathcal{R}$) and Doc. 204 (fractal boundary condition).

11 Refinement 8: Terminology fractal renormalization

Affected Documents

Doc. 005, 008, 012, 014 (with 19 occurrences the most frequent source), 041, 060, 133, 141, 159, 181.

Previous Formulation

Fractal renormalization is used as an FFGFT-internal term for the geometric correction factors in the conversion between natural and SI units, and for scale-dependent corrections on the fractal torus.

Refined Formulation

Since the word renormalization – even in compound form – carries the QFT connotation of a subtraction or adjustment procedure, the term is replaced by context-specific alternatives:

Context	**Binding designation**
Unit conversion natural $\leftrightarrow$ SI (esp. Doc. 014)	Geometric scale adjustment
Scale-dependent corrections on the fractal torus (Doc. 133, 159, 181)	Fractal correction (matches title of Doc. 133)
General scale transformation by recursion	$\mathcal{R}$-induced scale correction

Justification

The proprietary term fractal renormalization was meant to distinguish the procedure clearly from QFT renormalization, but it adopts the central word and unavoidably with it the connotation of a *correction procedure*. In FFGFT this is not a procedure but the direct numerical manifestation of the fractal geometry. The proposed alternatives avoid this misunderstanding without requiring new vocabulary – both phrases are already established in the corpus (Doc. 133 fractal correction; geometric scale adjustment in the same semantic field as Doc. 014).

Updated Summary

No.	Doc.	Erroneous / unclear	Correct / refined
C1	049 (HTML)	$\xi = \lambda_h^2 v^2/(64\pi^4 m_h^2)$	$\xi = \lambda_h^2 v^2/(16\pi^3 m_h^2)$
C2	116 (table)	$r_\tau = 8/3$	$r_\tau = 25/9$
C3	018 (Explorer)	$a_e = 4\pi/(f \cdot k)$	$a_e = 2\pi^2/(f \cdot k)$
R1	116 / 158	$\Delta Q < 0.00003\,\%$ vs. $0.001\,\%$	Both correct; $0.001\,\%$ for external use
R2	several	$\hbar$ as a postulate	$\hbar$ follows from ξ via $L_0 = \xi \cdot \ell_P$
R3	173–176	A single ξ value	Two parameters: ξ_0 (geom.) and ξ_{Higgs} (alg.)
R4	180–182	L_0 as Schwarzschild radius	L_0 = geom. fundamental length
R5	several	FFGFT calculates masses exactly	Mass structure from ξ; $\sim 1\,\%$ agreement
R6	116 / 158	Koide with FFGFT symbolic terms	Koide only with numerical values (non-closure, Doc. 193)
R7	005, 008, 019, 086, 189, 202	Renormalization group as FFGFT tool	Scale structure from recursion $\mathcal{R}$
R8	005, 008, 012, 014, 041, 060, 133, 141, 159, 181	Fractal renormalization as proprietary term	Geometric scale adjustment / fractal correction